SAEED TURABZADEH

Technical Lead | Autonomous Systems Expert | Real-Time 2D/3D Perception | Deep Learning | Scalable AI Solutions | Sensor Fusion | End-to-End Deployment | C++ / Python

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STRENGTHS

Technical Leadership

Led cross-functional engineering teams, driving complex autonomy projects from concept to deployment.

Collaboration

Collaborated with cross-functional global teams to deliver high-impact, scalable solutions.

Attention to Detail

Ensured precision in algorithm design, code quality, and system validation for robust autonomous performance.

Problem Solving

Applied innovative thinking to resolve perception, ML, and system integration challenges.

Client Engagement

Worked closely with international clients to capture requirements and deliver tailored autonomy systems.

INDUSTRY EXPERTISE

Technical Leadership & Delivery

Autonomous Systems Development

Real-Time Perception & Control

2D/3D Vision & Deep Learning

C++ & Python Programming

Sensor Fusion & Data Processing

AI & ML Deployment

AWARDS

- 2014, Winner of the prestigious VC-Chancellor's research prize, Brunel University London
- 2014, Diploma in Management (QCF), London

SUMMARY

Experienced and results-driven Technical Lead with a strong record of leading cross-functional engineering teams to deliver scalable, production-grade autonomous systems. Proven ability to define technical roadmaps, manage complex system integrations, and align engineering execution with business objectives. Skilled in stakeholder engagement, team mentoring, and driving full-cycle product development across international projects in the UK, UAE, and Southeast Asia.

Proficient in autonomous systems, perception, and applied machine learning, with deep expertise in real-time software development using C++ and Python. Specialized in 2D/3D computer vision, sensor fusion, and embedded systems for robotics and autonomous vehicles. Strong hands-on experience with system architecture, data pipelines, deep learning model deployment, and multi-platform integration. Passionate about solving real-world problems through cutting-edge AI and automation technologies.

EXPERIENCE

Technical Lead — 11/2021 - 05/2025

Aidrivers — London, UK

Led the development of scalable, real-time autonomous mobility solutions across multiple international projects (Singapore, Malaysia, Sri Lanka, UAE). Combined strategic oversight with hands-on engineering in perception, decision-making, and control systems.

- Directed software architecture, sensor integration, and multi-platform deployment, aligning technical execution with business goals.
- Designed end-to-end sensor setups, managed hardware selection, and optimized configurations to meet project-specific needs.
- Defined technical roadmaps, testing frameworks, and validation strategies, driving projects from concept to production.
- Acted as lead engineer and client liaison, translating operational requirements into actionable development plans and delivering robust, production-grade systems.

Senior Autonomy Engineer — 03/2020 - 10/2021

Aidrivers — London, UK

Contributed to the development of real-time perception and autonomy systems, focusing on 2D/3D computer vision, deep learning, and system reliability. Collaborated closely with cross-functional engineering teams to deliver production-ready solutions.

- Developed and optimized 2D deep learning models for lane detection, object detection, traffic light recognition, and environment classification. Converted models to ONNX and deployed TensorRT engines for real-time C++ integration.
- Enhanced 3D perception with Patchwork ground filtering and 2D-to-3D projection, improving object and lane detection accuracy with multi-modal tracking.
- Led annotation strategy and managed dataset creation workflows, ensuring high-quality labeled data and streamlined operations.
- Built an automated dataset backup system and introduced a fault-tolerant architecture to increase system reliability and availability.

Autonomy Engineer — 06/2018 - 02/2020

Aidrivers — London, UK

Worked on core autonomy features with a focus on computer vision and deep learning for automation systems. Contributed to the development and deployment of perception modules, supporting data-driven solutions in real-world environments.

- Implemented and optimized computer vision algorithms for object detection, classification, and tracking in dynamic scenes.
- Applied deep learning models to enhance perception accuracy and system robustness in autonomous workflows.
- Supported integration and testing of vision pipelines, contributing to the automation of decision-making and control systems.

SKILLS

Technical Leadership, Team Mentoring, Cross-Functional Collaboration

Strategic Roadmapping, Software Architecture Design, End-to-End Deployment

C++, Python, C#, Bash / Shell Scripting

2D Object Detection, Lane Detection, Classification, 3D Clustering, OCR, Sensor Fusion, 2D–3D Projection

PyTorch, TensorFlow, ONNX, TensorRT, OpenCV, PCL, Model Optimization

ROS / ROS2, Qt (C++/Python), PySide2, PyQt

Linux (Ubuntu), NVIDIA Jetson, CUDA, cuDNN, Real-Time Systems

Docker, Git/GitHub, CMake, CVAT, JIRA, Confluence

PUBLICATIONS

Facial Expression Emotion Detection for Real-Time Embedded Systems

MDPI - Technologies

2018
https://www.mdpi.com/257540

Real-time Emotional State Detection from Facial Expression on Embedded Devices

INTECH

2017
https://ieeexplore.ieee.org/document/8102423/

Automatic Affective Dimension Recognition from Naturalistic Facial Expressions Based on Wavelet Filtering and PLS Regression

IEEE

2015
https://ieeexplore.ieee.org/document/7284859/

Automatic Depression Scale Prediction using Facial Expression Dynamics and Regression, Proceedings of the 4th International Workshop on Audio/Visual Emotion Challenge (AVEC14)

AVEC

2014
https://dl.acm.org/citation.cfm?id=2661812

EXPERIENCE

Robotics Engineer — 08/2016 - 05/2018

DIDRIVERS — London, UK

Worked on real-time robotics and vision systems, focusing on computer vision, embedded communication, and hardware-software integration. Collaborated with international teams, including those in Dubai, on product development and deployment.

- Developed and implemented vision-based features such as face recognition, emotion detection, lane tracking, video summarization, and obstacle avoidance to improve autonomous performance and safety.
- Supported system integration, technical documentation, and product demonstrations, ensuring smooth communication between hardware and software components in robotics applications.

Assistant Systems Engineer — 09/2015 - 08/2016

Tata Consultancy Services — London, UK

Assisted in system design, testing, and documentation, contributing to improvements and ensuring technical requirements were met.

- Coordinated system design, integration, and formal testing activities across engineering teams.
- Created and maintained technical documentation, and proposed design improvements for existing systems.

IT Technical Engineer — 10/2015 - 05/2018

Delaware North — London, UK

Graduate Researcher & Teaching Assistant — 09/2013 - 08/2015

Brunel University London — London, UK

EDUCATION

MPhil, Electronic and Computer Engineering — 2013 - 2015

Brunel University London — London, UK

BEng, Electronic and Communication Engineering — 2010 - 2013

Brunel University London — London, UK

PROJECTS

Autonomous Terminal Trucks (APM)

Developed perception and decision-making modules for autonomous terminal trucks in seaport environments. Designed multi-LiDAR and camera-based systems for obstacle detection, traffic light recognition, lane detection, and navigation. Integrated real-time sensor fusion, deep learning perception, and control pipelines to enable safe and efficient automated port operations.

Chassis Alignment System (CAS)

Led the development of a combined LiDAR and camera solution for chassis alignment and container handover automation. Implemented real-time detection, alignment verification, and control algorithms to synchronize crane positioning with vehicle movements, improving safety and operational efficiency.

Ship Profile Scanning System (SPSS)

Developed a LiDAR and camera-based system for real-time vessel profiling, detecting container stacks and vessel structures to automate crane operation planning. Integrated multi-sensor data fusion and optimized point cloud processing for high-precision ship profiling.

Humanoid Robot (WIZO)

Built a multi-functional robot with embedded vision for real-time object and facial recognition. Integrated natural language processing and multilingual communication for user interaction, with video content analysis for AI-assisted engagement and memory-efficient recording.